# Lesson 65: Communication in Multi-Agent RL

AI/ML Mathematics

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# 1 Motivation and Overview

Consider a team of autonomous search-and-rescue drones deployed over a disaster zone. Each drone has a limited field of view and can only observe a small region of the terrain at any time step. When one drone spots a survivor, it cannot carry out the rescue alone—it needs teammates to converge on the location, perhaps bringing medical supplies or forming a relay to a base station. The challenge is immediate: the drone that detects the survivor must somehow *inform* its teammates, who are distributed across the zone, each processing only their own local observations. But what should the message contain? The raw observation? A compressed encoding? A directive? And to whom should the message be sent—all drones (broadcast) or only the nearest ones (targeted)?

This scenario exposes a fundamental tension in multi-agent reinforcement learning: decentralized agents operating under partial observability can, in principle, achieve near-centralized performance—*if* they communicate effectively. The tools developed in this lesson will let us formalize communication channels within the Dec-POMDP framework, analyze when communication provably improves team performance, and derive end-to-end learning algorithms for communication protocols.

**Why Communication Matters for RL/ML**

- **Robot swarm coordination.** Swarms of robots performing exploration, construction, or foraging rely on learned communication to coordinate without centralized control.
- **Autonomous vehicle negotiation.** Vehicles at intersections without traffic signals must exchange intentions to avoid collisions, requiring compact communication protocols.
- **Distributed sensor networks.** Bandwidth-limited sensors must decide what information to transmit to a fusion center, balancing informativeness against channel cost.
- **Emergent language in AI.** Studying how RL agents develop communication protocols from scratch connects to fundamental questions about the origins of language.

**Prerequisites.** Stochastic game formalism and Markov perfect equilibrium (Lesson 61, Definitions 3.1–3.4). Cooperative MARL, Dec-POMDP framework, and centralized training with decentralized execution (Lesson 62, Sections 3–5). Policy gradient theorem and REINFORCE (Lessons 54–55). Neural network forward and backward propagation (Phase 03, Lessons 59–66).

**Learning objectives.** After completing this lesson, you will be able to:

- **Define** a communication-augmented Dec-POMDP and **classify** communication protocols by channel type, topology, and learning method.
- **Derive** the CommNet gradient and **explain** why end-to-end backpropagation through continuous communication is well-defined.
- **Compare** DIAL and RIAL for discrete communication, identifying the variance trade-off between reparameterization and REINFORCE-style gradients.
- **Formalize** TarMAC's attention-based targeted communication mechanism and **prove** that it subsumes CommNet as a special case.
- **Analyze** conditions under which emergent communication arises and **evaluate** metrics for compositionality.
- **Justify** when communication improves team performance over non-communicative baselines using the value of communication framework.

## 2 References

- Sukhbaatar et al. [2016] — CommNet: continuous communication via mean-pooled hidden states with end-to-end backpropagation.
- Foerster et al. [2016] — DIAL and RIAL: differentiable and reinforced approaches to discrete inter-agent communication.
- Das et al. [2019] — TarMAC: targeted multi-agent communication using soft attention over agent messages.
- Zhang et al. [2021] — Comprehensive survey of multi-agent RL theory and algorithms, including communication methods.
- Hernandez-Leal et al. [2019] — Critical survey of deep MARL covering communication architectures and open problems.

## 3 Communication-Augmented Dec-POMDP

We begin by extending the standard Dec-POMDP framework of Lesson 62 with explicit communication channels. The key idea is that agents can exchange messages before (or simultaneously with) selecting actions, and these messages become part of the information available for decision-making.

**Definition 3.1** (Message Space)**.** For a multi-agent system with agent set $\mathcal{N} = \{1, \dots, N\}$, the *message space* of agent $i$ is a set $\mathcal{M}_i$. A message $m_i \in \mathcal{M}_i$ is a signal that agent $i$ can transmit at each time step. The message may be continuous ($\mathcal{M}_i \subseteq \mathbb{R}^d$) or discrete ($\mathcal{M}_i = \{1, \dots, M\}$).

**Definition 3.2** (Communication-Augmented Dec-POMDP)**.** A *communication-augmented Dec-POMDP* (Comm-Dec-POMDP) is a tuple

$$\mathcal{G} = \big(\mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}_{i\in\mathcal{N}}, \mathcal{P}, \{r_i\}_{i\in\mathcal{N}}, \{\Omega_i\}_{i\in\mathcal{N}}, \{O_i\}_{i\in\mathcal{N}}, \{\mathcal{M}_i\}_{i\in\mathcal{N}}, \mathcal{C}, \gamma\big),$$

where $\mathcal{S}$ is the state space, $\mathcal{A}_i$ is agent $i$'s action space, $\mathcal{P}$ is the transition kernel, $r_i$ is agent $i$'s reward, $\Omega_i$ is agent $i$'s observation space, $O_i : \mathcal{S} \to \Delta(\Omega_i)$ is the observation function, $\mathcal{M}_i$ is the message space, $\mathcal{C}$ is the *communication protocol*, and $\gamma \in [0, 1)$ is the discount factor.

The communication protocol $\mathcal{C}$ specifies:

1. **Timing:** whether messages are sent before or simultaneously with actions;
2. **Topology:** a directed graph $G = (\mathcal{N}, E)$ where $(i, j) \in E$ means agent $i$ can send messages to agent $j$;
3. **Bandwidth:** constraints on $|\mathcal{M}_i|$ or $\dim(\mathcal{M}_i)$.

Agent $i$'s policy maps its local observation and received messages to a joint action-message output:

$$\pi_i : \Omega_i \times \prod_{j:(j,i)\in E} \mathcal{M}_j \longrightarrow \Delta(\mathcal{A}_i \times \mathcal{M}_i).$$

*Remark* 3.3 (Reduction to Centralized POMDP). Consider a Comm-Dec-POMDP with broadcast topology ($E = \mathcal{N} \times \mathcal{N} \setminus \{(i, i)\}$) and message spaces satisfying $|\mathcal{M}_i| \geq |\Omega_i|$ for all $i$. Then each agent can transmit its full observation, and after one round of communication every agent has access to the joint observation $(o_1, \dots, o_N)$. The resulting information structure is equivalent to a centralized POMDP.

**Proposition 3.4** (Communication Reduces Complexity)**.** *The optimal finite-horizon Dec-POMDP problem is NEXP-complete [Busoniu et al., 2008]. If communication channels satisfy $|\mathcal{M}_i| \geq |\Omega_i|$ for all $i \in \mathcal{N}$ with broadcast topology, then the Comm-Dec-POMDP reduces to a centralized POMDP, which is PSPACE-complete.*

*Proof.* Under broadcast communication with $|\mathcal{M}_i| \geq |\Omega_i|$, agent $i$ can encode its observation $o_i$ into a message $m_i = \text{enc}(o_i)$ with a bijective encoding. After one communication round, every agent $j$ receives $\{m_i\}_{i \neq j}$ and can decode to obtain $\{o_i\}_{i \neq j}$. Combined with its own observation $o_j$, agent $j$ has access to the full joint observation $(o_1, \ldots, o_N)$.

When all agents observe the same information, the joint policy optimization can be cast as a single-agent POMDP with state space $\mathcal{S}$, action space $\mathcal{A}_1 \times \cdots \times \mathcal{A}_N$, and observation space $\Omega_1 \times \cdots \times \Omega_N$. The finite-horizon POMDP is PSPACE-complete (Papadimitriou and Tsitsiklis, 1987), so the Comm-Dec-POMDP inherits this complexity class under sufficient communication. □

The following taxonomy organizes the main axes along which communication protocols differ.

**Definition 3.5** (Communication Taxonomy)**.** Communication protocols in MARL are classified along three axes:

1. **Channel type:** *Continuous* ($\mathcal{M}_i \subseteq \mathbb{R}^d$) vs. *discrete* ($\mathcal{M}_i$ finite).
2. **Topology:** *Broadcast* (all-to-all) vs. *targeted* (agent-specific recipients) vs. *bandwidth-limited* (fixed number of recipients).
3. **Learning method:** *End-to-end differentiable* vs. *reinforcement-based* vs. *pre-defined* (hand-crafted protocol).

**Bridging Centralized and Decentralized Performance**

Communication is the mechanism that interpolates between two extremes: fully decentralized execution (no communication, each agent acts on local observations alone) and fully centralized control (a single controller with access to all observations). The communication bandwidth determines where in this spectrum the system operates. In practice, even a small amount of learned communication can dramatically close the gap between decentralized and centralized performance [Sukhbaatar et al., 2016].

We now turn to the three major algorithmic paradigms for learning communication protocols: continuous communication (Section 4), discrete communication (Section 5), and targeted communication (Section 6).

# 4 Continuous Communication: CommNet

CommNet [Sukhbaatar et al., 2016] is the canonical continuous communication architecture. All agents share a single neural network, and messages are the hidden states of other agents, averaged and fed back as input.

## 4.1 Architecture

Consider $N$ agents with shared parameters $\theta$. At each time step $t$, the CommNet computation proceeds in $K$ communication rounds.

**Definition 4.1** (CommNet)**.** The CommNet architecture with $K$ communication rounds is defined by the following recurrence. For each agent $i \in \mathcal{N}$:

1. **Encoding:** $\mathbf{h}_i^{(0)} = f_{\text{enc}}(o_i; \theta_{\text{enc}})$, where $f_{\text{enc}} : \Omega_i \to \mathbb{R}^d$.
2. **Communication rounds** ($k = 1, \ldots, K$):

$$\mathbf{c}_i^{(k)} = \frac{1}{N-1} \sum_{j \neq i} \mathbf{h}_j^{(k-1)}, \tag{1}$$

$$\mathbf{h}_i^{(k)} = \sigma\big(W_1^{(k)} \mathbf{h}_i^{(k-1)} + W_2^{(k)} \mathbf{c}_i^{(k)}\big), \tag{2}$$

where $W_1^{(k)}, W_2^{(k)} \in \mathbb{R}^{d \times d}$ are learnable weight matrices and $\sigma$ is an element-wise nonlinearity (e.g., ReLU or tanh).

3. **Decoding:** $\pi_i(\cdot \mid o_i, \mathbf{c}_i) = \mathrm{softmax}\big(f_{\mathrm{dec}}(\mathbf{h}_i^{(K)}; \theta_{\mathrm{dec}})\big)$.

The "message" in CommNet is the mean of other agents' hidden states—a continuous vector in $\mathbb{R}^d$. This design ensures that the entire computation graph, from observations to actions, is a composition of differentiable operations.

### 4.2 End-to-End Differentiability

**Theorem 4.2** (CommNet Gradient)**.** *Let $J(\theta) = \mathbb{E}\big[\sum_{t=0}^{T} \gamma^t \sum_{i \in \mathcal{N}} r_i(s_t, \mathbf{a}_t)\big]$ be the team objective under CommNet with parameters $\theta = \{\theta_{\mathrm{enc}}, \{W_1^{(k)}, W_2^{(k)}\}_{k=1}^{K}, \theta_{\mathrm{dec}}\}$. Then $\nabla_\theta J(\theta)$ is well-defined and can be computed by standard backpropagation through the communication rounds.*

*Proof.* We show that every operation in the CommNet computation graph is differentiable with respect to $\theta$.

**Step 1: Encoding.** The encoder $f_{\mathrm{enc}}$ is a neural network with differentiable activation functions, so $\partial \mathbf{h}_i^{(0)} / \partial \theta_{\mathrm{enc}}$ exists.

**Step 2: Communication rounds.** Fix round $k$. The communication message (1) is a linear combination (mean) of the hidden states $\{\mathbf{h}_j^{(k-1)}\}_{j \neq i}$, which is differentiable. The hidden state update (2) is a composition of an affine map $(W_1^{(k)}, W_2^{(k)})$ and the nonlinearity $\sigma$. If $\sigma$ is differentiable almost everywhere (as with ReLU or tanh), then $\mathbf{h}_i^{(k)}$ is differentiable with respect to both $\mathbf{h}_i^{(k-1)}$ and the parameters $W_1^{(k)}, W_2^{(k)}$.

**Step 3: Induction.** By induction on $k$, each $\mathbf{h}_i^{(k)}$ is differentiable with respect to $\theta$. The key observation is that the mean operation in (1) couples all agents: $\partial \mathbf{c}_i^{(k)} / \partial \mathbf{h}_j^{(k-1)} = \frac{1}{N-1} I_d$ for $j \neq i$. This coupling propagates gradients across agents.

**Step 4: Decoding and objective.** The decoder $f_{\mathrm{dec}}$ and softmax are differentiable. For deterministic action selection, the gradient flows directly. For stochastic policies, we apply the reparameterization trick (for continuous actions) or the log-derivative trick combined with the fact that the logits are differentiable in $\theta$.

The overall gradient $\nabla_\theta J$ is obtained by backpropagating through the temporal sequence and through the communication rounds at each time step. Since the composition of differentiable functions is differentiable, the gradient is well-defined. □

**End-to-End Differentiable Communication**

CommNet's use of continuous messages (mean-pooled hidden states) makes the entire multi-agent system differentiable with respect to shared parameters. Standard backpropagation computes the gradient without requiring REINFORCE-style estimators for the communication component, yielding low-variance gradient estimates.

### 4.3 Properties and Limitations

**Example 4.3** (Three-Agent Hallway Navigation)**.** Consider three agents in a hallway with three rooms. Each agent observes only its current room. The team receives reward $+1$ when all agents are in different rooms at the same time step. Without communication, agents must resort to randomized strategies.

With CommNet ($K = 1$), each agent broadcasts its encoded observation. The received message $\mathbf{c}_i^{(1)} = \frac{1}{2}(\mathbf{h}_j^{(0)} + \mathbf{h}_k^{(0)})$ encodes the average position of the other two agents. This suffices for each agent to infer a "free" room and navigate to it, achieving near-optimal coordination.

*Remark* 4.4 (Limitations of Mean-Pooled Communication). CommNet has two principal limitations:

1. **No targeted messaging.** Every agent receives the same aggregate message (the mean of all others' hidden states), so agent-specific information is lost.
2. **Permutation symmetry.** The mean operation is invariant to permutations of agents, which can be beneficial (scalability) but also limits expressiveness when agent identities matter.

These limitations motivate the targeted communication mechanisms of Section 6.

# 5 Discrete Communication: DIAL and RIAL

In many real-world deployments, communication channels carry discrete signals: binary flags, categorical indices, or natural language tokens. The fundamental challenge is that discrete message selection (e.g., $\arg\max$) is not differentiable, which breaks the end-to-end gradient path exploited by CommNet. Foerster et al. [2016] proposed two approaches: RIAL (reinforced) and DIAL (differentiable).

## 5.1 RIAL: Reinforced Inter-Agent Learning

**Definition 5.1** (RIAL Policy)**.** In RIAL, agent $i$'s policy produces both an environment action and a discrete message:

$$\pi_i(a_i, m_i \mid o_i, \mathbf{m}_{-i}^{\text{recv}}) = \pi_i^a(a_i \mid o_i, \mathbf{m}_{-i}^{\text{recv}}) \;\cdot\; \pi_i^m(m_i \mid o_i, \mathbf{m}_{-i}^{\text{recv}}),$$

where $m_i \in \{1, \ldots, M\}$ is a discrete message, $\mathbf{m}_{-i}^{\text{recv}}$ denotes the messages received from other agents, and the action and message components may share a common hidden representation.

In RIAL, the message $m_i$ is treated as an additional action. The gradient for the message component is estimated via the policy gradient (REINFORCE):

$$\nabla_\theta J^{\text{RIAL}} = \mathbb{E}\Big[\sum_{i \in \mathcal{N}} \nabla_\theta \log \pi_i(a_i, m_i \mid \cdot\,; \theta)\; \hat{A}_i\Big], \tag{3}$$

where $\hat{A}_i$ is an advantage estimate. Crucially, there is *no gradient flowing through the communication channel*: agent $j$ treats the received message $m_i$ as part of the environment, not as a function of $\theta$.

## 5.2 DIAL: Differentiable Inter-Agent Learning

**Definition 5.2** (Discretize/Regularize Unit (DRU))**.** The DRU maps a continuous pre-message $\tilde{m} \in \mathbb{R}^d$ to a channel-compatible signal:

$$\text{DRU}_\tau(\tilde{m}) = \begin{cases} \sigma(\tilde{m}/\tau), & \text{during training (temperature } \tau > 0), \\ \mathbf{1}[\tilde{m} > 0], & \text{during execution (hard discretization)}, \end{cases}$$

where $\sigma$ denotes the logistic sigmoid applied element-wise, $\tau$ is a temperature parameter, and $\mathbf{1}[\cdot > 0]$ is the element-wise indicator function.

**Definition 5.3** (DIAL)**.** In DIAL, agent $i$ produces a continuous pre-message $\tilde{m}_i = g_i(o_i, \mathbf{m}_{-i}^{\text{recv}}; \theta)$ via a neural network. During training, the message transmitted is $m_i = \text{DRU}_\tau(\tilde{m}_i)$, which is differentiable in $\theta$. During execution, the message is discretized: $m_i = \mathbf{1}[\tilde{m}_i > 0]$.

The DIAL gradient for the communication component is:

$$\frac{\partial J}{\partial \theta} \;\ni\; \frac{\partial J}{\partial a_j} \frac{\partial a_j}{\partial m_i} \frac{\partial m_i}{\partial \tilde{m}_i} \frac{\partial \tilde{m}_i}{\partial \theta}, \tag{4}$$

where $\partial m_i / \partial \tilde{m}_i = \sigma'(\tilde{m}_i/\tau)/\tau$ during training. This is a reparameterization-like gradient that passes through the communication channel.

### 5.3 Variance Comparison

**Proposition 5.4** (DIAL Has Lower Variance Than RIAL)**.** *For the communication component of the gradient, DIAL provides lower-variance estimates than RIAL. Specifically, let* $g^{\text{DIAL}}$ *and* $g^{\text{RIAL}}$ *denote the gradient estimators for the message parameters. Then*

$$\text{Var}[g^{\text{DIAL}}] \leq \text{Var}[g^{\text{RIAL}}].$$

*Proof.* The RIAL gradient estimator for the message component has the REINFORCE form:

$$g^{\text{RIAL}} = \nabla_\theta \log \pi_i^m(m_i \mid \cdot\,; \theta) \;\cdot\; \hat{A}_i.$$

This is the product of the score function and the advantage, and its variance includes the term $\text{Var}[\hat{A}_i] \cdot \mathbb{E}[\|\nabla_\theta \log \pi_i^m\|^2]$ plus cross-terms.

The DIAL gradient estimator uses the pathwise derivative:

$$g^{\text{DIAL}} = \frac{\partial J}{\partial m_i} \cdot \frac{\partial \text{DRU}_\tau(\tilde{m}_i)}{\partial \tilde{m}_i} \cdot \frac{\partial \tilde{m}_i}{\partial \theta}.$$

Each factor is computed by backpropagation through deterministic operations (given the sampled trajectory). The variance arises only from the stochasticity of the trajectory, not from a log-probability score function.

By the general result that pathwise (reparameterization) estimators have lower variance than score-function (REINFORCE) estimators when both are applicable [Sutton and Barto, 2018], we conclude $\text{Var}[g^{\text{DIAL}}] \leq \text{Var}[g^{\text{RIAL}}]$. □

### 5.4 Comparison Summary

| | Gradient type | Variance | Channel | Execution |
|---|---|---|---|---|
| RIAL | REINFORCE (score) | High | Discrete | Discrete |
| DIAL | Pathwise (reparam.) | Low | Continuous (train) | Discrete |

**Warning**

During training, DIAL uses the continuous relaxation $\sigma(\tilde{m}/\tau)$, but during execution, messages are discretized to $\mathbf{1}[\tilde{m} > 0]$. If the temperature $\tau$ is not annealed toward zero during training, the learned policy may depend on continuous message values that are lost upon discretization, causing performance degradation at test time. Careful scheduling of $\tau \to 0$ is essential.

**Example 5.5** (Two-Agent Switch Riddle)**.** Two agents are placed in separate rooms, each containing a switch (up or down). Agent 1 observes the state of both switches but can only flip switch 1. Agent 2 can flip switch 2 but observes nothing. The team receives reward +1 if both switches end up in the same position.

Without communication, agent 2 must guess. With a 1-bit communication channel ($m_1 \in \{0, 1\}$), agent 1 can signal the state of switch 2, and agent 2 can act optimally. RIAL learns this by reinforcing the correct message; DIAL learns it by backpropagating the reward signal through the sigmoid-relaxed channel. In experiments [Foerster et al., 2016], DIAL converges faster due to its lower-variance gradient.

## 6 Targeted Communication: TarMAC

CommNet broadcasts a uniform average to all agents, ignoring the fact that different agents may need different information. TarMAC [Das et al., 2019] introduces an attention mechanism that allows agents to send targeted messages, weighted by relevance.

### 6.1 Architecture

**Definition 6.1** (TarMAC)**.** In TarMAC, each agent $i$ produces three quantities from its hidden state $\mathbf{h}_i$:

$$\mathbf{v}_i = f_v(\mathbf{h}_i; \theta_v) \in \mathbb{R}^{d_v}, \qquad \text{(value/message)}, \tag{5}$$

$$\mathbf{k}_i = f_k(\mathbf{h}_i; \theta_k) \in \mathbb{R}^{d_k}, \qquad \text{(key/signature)}, \tag{6}$$

$$\mathbf{q}_i = f_q(\mathbf{h}_i; \theta_q) \in \mathbb{R}^{d_k}, \qquad \text{(query)}. \tag{7}$$

The attention weight from agent $i$ to agent $j$ is:

$$\alpha_{j \leftarrow i} = \frac{\exp(\mathbf{q}_j^\top \mathbf{k}_i / \sqrt{d_k})}{\sum_{l \neq j} \exp(\mathbf{q}_j^\top \mathbf{k}_l / \sqrt{d_k})}, \tag{8}$$

and the aggregated message received by agent $j$ is:

$$\mathbf{c}_j = \sum_{i \neq j} \alpha_{j \leftarrow i} \, \mathbf{v}_i. \tag{9}$$

Agent $j$ then conditions its policy on $(\mathbf{h}_j, \mathbf{c}_j)$.

This is precisely the scaled dot-product attention mechanism from transformer architectures, applied across agents rather than across sequence positions. Multi-head variants extend this by computing $H$ independent attention heads and concatenating the results.

### 6.2 Subsumption of CommNet

**Proposition 6.2** (TarMAC Subsumes CommNet)**.** *If the attention weights in TarMAC are uniform, i.e., $\alpha_{j \leftarrow i} = 1/(N-1)$ for all $i \neq j$, and the value function is the identity $\mathbf{v}_i = \mathbf{h}_i$, then TarMAC reduces to CommNet's mean communication.*

*Proof.* Under uniform attention, the aggregated message (9) becomes

$$\mathbf{c}_j = \sum_{i \neq j} \frac{1}{N-1} \, \mathbf{v}_i = \frac{1}{N-1} \sum_{i \neq j} \mathbf{h}_i,$$

where the last equality uses $\mathbf{v}_i = \mathbf{h}_i$. This is exactly the CommNet communication (1). □

**Attention-Based Targeted Communication**

TarMAC's attention mechanism allows each agent to selectively attend to the most relevant messages, generalizing CommNet's uniform broadcast. The attention weights are learned end-to-end, enabling the system to discover which agents should communicate with which. As a strict generalization of CommNet, TarMAC achieves at least the same representational power.

**Attention and Scalability**

The attention-based communication in TarMAC mirrors the self-attention mechanism in transformers. This connection suggests that advances in efficient attention (sparse attention, linear attention) can be directly applied to scale multi-agent communication to large agent populations. Furthermore, attention weights provide interpretability: examining $\alpha_{j \leftarrow i}$ reveals which agents are communicating most, aiding in debugging and understanding emergent strategies.

**Example 6.3** (Predator-Prey with Targeted Communication)**.** Consider four predators pursuing a single prey in a grid world. Each predator observes a $5 \times 5$ patch centered on itself. When a predator is near the prey but needs help to capture it (capture requires two predators adjacent to the prey simultaneously), it should communicate primarily with the nearest other predator, not all of them.

With TarMAC, the predator near the prey learns to produce a high-attention key $\mathbf{k}_i$ that matches the query $\mathbf{q}_j$ of the nearest predator $j$. The attention weight $\alpha_{j \leftarrow i}$ becomes large, routing the positional information to the most relevant teammate. Distant predators receive low attention weight and are not distracted by irrelevant messages.

# 7 Emergent Communication and Language Grounding

Sections 4–6 treated communication as a mechanism for improving team performance. A deeper question is whether agents can develop a structured *language* from scratch—a communication protocol with compositional, interpretable structure.

**Definition 7.1** (Emergent Communication)**.** *Emergent communication* refers to a communication protocol that arises purely from multi-agent RL training, without any pre-defined language, message semantics, or explicit supervision on the communication channel. The protocol is shaped solely by the reward signal.

## 7.1 Referential Games

The standard testbed for emergent communication is the *referential game* [Lazaridou et al., 2017].

**Definition 7.2** (Referential Game)**.** A referential game consists of:

- A *speaker* agent $S$ that observes a target object $x \in \mathcal{X}$ (e.g., an image);
- A message space $\mathcal{M}$ (typically a sequence of discrete symbols);
- A *listener* agent $L$ that receives the message $m$ and must identify $x$ from a set of candidates $\{x_1, \ldots, x_C\}$;
- Reward: $r = \mathbf{1}[L \text{ selects the correct } x]$.

## 7.2 Compositionality

**Definition 7.3** (Compositionality)**.** An emergent language is *compositional* if it exhibits systematic structure: the meaning of a complex message is determined by the meanings of its parts and the rules for combining them. Formally, let $\mathcal{X}$ be a structured input space (e.g., objects described by attributes such as color, shape, size) and let $\phi : \mathcal{X} \to \mathcal{M}$ be the speaker's encoding. The language is compositional if there exists a decomposition

$$\phi(x) = (\phi_1(\text{attr}_1(x)),\ \phi_2(\text{attr}_2(x)),\ \ldots)$$

where each $\phi_k$ maps the $k$-th attribute independently to a message component.

**Definition 7.4** (Topographic Similarity)**.** The *topographic similarity* (topo-sim) of a communication protocol $\phi$ is the Spearman correlation between distances in meaning space and distances in message space:

$$\rho_{\text{topo}} = \text{Spearman}\big(\{d_{\mathcal{X}}(x, x')\}_{x \neq x'},\ \{d_{\mathcal{M}}(\phi(x), \phi(x'))\}_{x \neq x'}\big),$$

where $d_{\mathcal{X}}$ and $d_{\mathcal{M}}$ are distance metrics on the input and message spaces, respectively. A topo-sim close to $+1$ indicates that semantically similar inputs are mapped to similar messages (compositionality).

### 7.3 Conditions for Emergence

**Theorem 7.5** (Conditions for Non-Trivial Communication)**.** *In a cooperative multi-agent setting with aligned incentives, non-trivial emergent communication (i.e., communication that improves team reward over non-communicative baselines) arises when the following conditions hold:*

1. ***Partial observability:*** *no single agent observes the full state;*
2. ***Complementary information:*** *different agents have access to different relevant features of the state;*
3. ***Action interdependence:*** *the optimal action for at least one agent depends on information held by another;*
4. ***Sufficient channel capacity:*** $|\mathcal{M}_i|$ *is large enough to encode the relevant information.*

*Proof.* We prove this by establishing necessity.

**Necessity of (1):** If all agents observe the full state $s$, then each agent can independently compute the optimal action $a_i^*(s)$ without any communication. Hence VoC $= 0$ (see Definition 8.1).

**Necessity of (2):** If agents have identical observations ($o_i = o_j$ for all $i, j$), then no agent possesses information that others lack. Any message is redundant since the receiver already has the same information as the sender.

**Necessity of (3):** If agent $i$'s optimal action does not depend on the observations of any other agent—i.e., $a_i^*(o_i) = \arg\max_{a_i} \mathbb{E}[r_i \mid o_i, a_i]$ regardless of $a_{-i}$ and $o_{-i}$—then communication cannot improve $i$'s policy. If this holds for all agents, VoC $= 0$.

**Necessity of (4):** If the channel capacity is zero ($|\mathcal{M}_i| = 1$), no information can be transmitted. More generally, the channel must support encoding of the relevant information with sufficient fidelity.

**Sufficiency:** When all four conditions hold, there exists at least one state $s$ and pair of agents $(i, j)$ such that agent $j$ can improve its action by receiving information from agent $i$ that is not available in $o_j$. A communication protocol mapping $o_i$ to a message $m_i$ that encodes this information yields strictly positive value of communication. □

**Warning**

Emergent communication protocols are typically not interpretable by humans. Agents may exploit artifacts of the channel (e.g., encoding information in the *timing* or *magnitude* of messages) rather than developing a human-readable language. Compositionality, when it emerges, often requires explicit pressure such as limited channel bandwidth or structured input distributions [Lazaridou et al., 2017].

**Emergent Communication and Language Evolution**

The study of emergent communication in MARL connects to deep questions in linguistics and cognitive science about how human languages evolved. Referential games serve as simplified models of the naming game studied in evolutionary linguistics. Computational experiments have shown that compositionality—a hallmark of human language—can emerge under appropriate pressures (limited bandwidth, generalization requirements), bridging MARL and computational linguistics.

## 8 Theoretical Analysis of Communication Benefits

We now formalize the question: *when does communication provably help?*

**Definition 8.1** (Value of Communication)**.** For a Dec-POMDP $\mathcal{G}$, let $V^*_{\text{comm}}$ denote the optimal team value achievable with a communication channel $\mathcal{C}$, and let $V^*_{\text{no-comm}}$ denote the optimal team value without communication. The *value of communication* is

$$\text{VoC}(\mathcal{C}) = V^*_{\text{comm}} - V^*_{\text{no-comm}} \;\geq\; 0.$$

The non-negativity VoC $\geq 0$ follows because any non-communicative policy is a special case of a communicative policy (with all messages set to a constant).

**Proposition 8.2** (Zero Value of Communication)**.** VoC$(\mathcal{C}) = 0$ *if there exists, for each agent $i$, a sufficient statistic $T_i(o_i)$ for the optimal action $a_i^*$ that depends only on $o_i$. Formally, if*

$$\pi_i^*(a_i \mid o_i) = \pi_i^*(a_i \mid o_i, m_{-i}) \quad \textit{for all } m_{-i} \in \prod_{j \neq i} \mathcal{M}_j,$$

*for every agent $i$, then communication has zero value.*

*Proof.* If agent $i$'s optimal policy is invariant to received messages for every $i$, then the optimal communicative policy can be achieved by a non-communicative policy (simply ignore all messages). Hence $V^*_{\text{comm}} = V^*_{\text{no-comm}}$. □

**Proposition 8.3** (Positive Value of Communication)**.** VoC$(\mathcal{C}) > 0$ *if and only if the following three conditions hold simultaneously:*

1. ***Partial observability:*** *there exists a state $s \in \mathcal{S}$ and agent $i$ such that $o_i$ does not determine $s$;*
2. ***Complementary information:*** *there exist agents $i, j$ and states $s, s'$ such that $o_i(s) \neq o_i(s')$ but $o_j(s) = o_j(s')$ (agent $i$ distinguishes states that agent $j$ cannot);*
3. ***Action interdependence:*** *agent $j$'s optimal action differs between $s$ and $s'$: $a_j^*(s) \neq a_j^*(s')$.*

*Proof.* **(If):** Under conditions (1)–(3), agent $j$ faces states $s, s'$ that it cannot distinguish ($o_j(s) = o_j(s')$) but that require different optimal actions. Without communication, $j$ must choose a single action for both states. With communication, agent $i$ (who can distinguish $s$ from $s'$) sends a message encoding this distinction. Agent $j$ conditions on the message and selects the state-appropriate action. The expected reward strictly increases, so VoC $> 0$.

**(Only if):** If (1) fails, all agents observe the full state and communication is redundant. If (2) fails, no agent has information that another lacks. If (3) fails, the optimal action is the same in all states that only communication could distinguish, so the additional information is useless. □

**Theorem 8.4** (Communication Complexity Bound)**.** *Consider a cooperative Comm-Dec-POMDP with $N$ agents, finite state space $|\mathcal{S}|$, and horizon $T$. An $\epsilon$-optimal joint policy (achieving value within $\epsilon$ of $V^*_{\text{comm}}$) can be achieved with at most*

$$B = O\left(|\mathcal{S}| \log \frac{T}{\epsilon}\right)$$

*bits of communication per agent per time step, assuming broadcast topology.*

*Proof sketch.* The key insight is that the relevant information for optimal decision-making is the sufficient statistic for the joint belief state.

**Step 1: Belief state representation.** The joint belief $b_t \in \Delta(\mathcal{S})$ requires at most $|\mathcal{S}| - 1$ real numbers to specify. Quantizing each coordinate to $\epsilon/(T \cdot |\mathcal{S}|)$ precision (to ensure total error over $T$ steps is bounded by $\epsilon$) requires $\log(T \cdot |\mathcal{S}|/\epsilon)$ bits per coordinate.

**Step 2: Sufficient communication.** If each agent broadcasts its local belief update (contribution to the joint belief), the team can reconstruct the joint belief. Each agent's contribution involves at most $|\mathcal{S}|$ coordinates, each requiring $O(\log(T/\epsilon))$ bits.

**Step 3: Total.** The total communication per agent per step is $O(|\mathcal{S}|\log(T/\epsilon))$ bits. With this precision, the team can implement a policy that is $\epsilon$-close to the optimal centralized policy. □

**Conditions for Communication Value**

The value of communication is positive if and only if agents have complementary partial observations and interdependent optimal actions. The communication bandwidth required for $\epsilon$-optimal performance scales as $O(|\mathcal{S}|\log(T/\epsilon))$ bits per agent per step—logarithmic in the horizon and the accuracy requirement, but linear in the state space size.

**Example 8.5** (VoC in the Switch Riddle)**.** Return to the switch riddle of Example 5.5. The state space is $\mathcal{S} = \{(\text{up}, \text{up}), (\text{up}, \text{down}), (\text{down}, \text{up}), (\text{down}, \text{down})\}$.

**Without communication:** Agent 2 observes nothing, so its policy must be independent of the state. The best strategy achieves reward 1/2 in expectation (guess randomly whether to flip).

**With 1-bit communication:** Agent 1 sends $m_1 = \mathbf{1}[\text{switch 2 is up}]$. Agent 2 conditions on $m_1$ and acts optimally, achieving reward 1. Hence $\text{VoC} = 1 - 1/2 = 1/2 > 0$.

All three conditions of Proposition 8.3 hold: (1) agent 2 has no observation; (2) agent 1 observes what agent 2 cannot; (3) agent 2's optimal action differs based on the unobserved switch state.

**Example 8.6** (VoC = 0 in Independent Tasks)**.** Consider $N$ agents each solving an independent bandit problem: agent $i$ observes its own bandit state $o_i$ and selects arm $a_i$. The team reward is $\sum_i r_i(o_i, a_i)$ with no interaction. Since each $r_i$ depends only on $(o_i, a_i)$, the optimal policy for agent $i$ is $a_i^*(o_i)$ regardless of other agents' observations. Condition (3) of Proposition 8.3 fails, so $\text{VoC} = 0$.

# 9 Survey and Comparison of Methods

We conclude the technical development with a comparative overview of the major communication methods in MARL.

## 9.1 Summary Table

| Method | Channel | Topology | Gradient | Scalability | Key Innovation |
|---|---|---|---|---|---|
| CommNet | Continuous | Broadcast | Backprop | Moderate | Mean-pooled hidden states |
| RIAL | Discrete | Broadcast | REINFORCE | Moderate | Message as RL action |
| DIAL | Discrete (exec.) | Broadcast | Backprop (train) | Moderate | DRU continuous relaxation |
| TarMAC | Continuous | Targeted | Backprop | Good | Attention over messages |
| IC3Net | Continuous | Gated | Backprop | Good | Gating when to communicate |
| ATOC | Continuous | Dynamic | Backprop | Good | Attention + dynamic groups |

IC3Net [Singh et al., 2019] extends CommNet with a learned gating mechanism that decides *when* to communicate, reducing unnecessary bandwidth usage. ATOC [Jiang and Lu, 2018] uses an attention mechanism to form dynamic communication groups based on agent proximity and relevance.

### 9.2 Method Selection Guidelines

- **CommNet:** Appropriate for simple cooperative tasks with a moderate number of agents ($N \leq 20$) where all agents benefit from a shared summary of the group state.
- **RIAL:** When the execution channel is strictly discrete and training-execution consistency is paramount. Accepts higher gradient variance in exchange for no train-test gap.
- **DIAL:** When discrete execution is required but sample efficiency is critical. The continuous relaxation during training significantly accelerates learning.
- **TarMAC:** Large agent populations ($N \geq 10$) where agents have heterogeneous roles and targeted communication matters. The attention mechanism provides natural scalability and interpretability.
- **IC3Net:** Mixed cooperative-competitive settings where agents should selectively withhold communication from opponents.

### 9.3 Open Problems

Several fundamental challenges remain:

1. **Scalability to thousands of agents.** Current methods scale to tens or hundreds of agents. Mean-field approximations (Lesson 64) may help, but integrating them with learned communication is largely unexplored.
2. **Adversarial robustness.** If agents can intercept or corrupt messages, communication protocols must be robust. Connections to Byzantine fault tolerance and robust game theory are open.
3. **Communication cost.** Most methods assume free communication. Incorporating explicit bandwidth costs into the objective function changes the optimization landscape significantly.
4. **Interpretability.** Developing communication protocols that are both performant and human-interpretable remains a grand challenge connecting MARL to natural language processing.

## 10 Exercises

**Exercise 10.1** (CommNet Mean Observation). ($*$)
Consider a CommNet with $K = 1$ communication round and identity encoder ($f_{\text{enc}}(o_i) = o_i$ with observations $o_i \in \mathbb{R}^d$). Show that if $W_1^{(1)} = 0$ and $W_2^{(1)} = I_d$ (identity), and $\sigma$ is the identity function, then each agent's hidden state after one round equals the mean of all *other* agents' observations: $\mathbf{h}_i^{(1)} = \frac{1}{N-1} \sum_{j \neq i} o_j$.

**Exercise 10.2** (RIAL Policy Gradient). ($*$)
Consider two agents ($\mathcal{N} = \{1, 2\}$) with binary message space $\mathcal{M}_i = \{0, 1\}$ and binary action space $\mathcal{A}_i = \{0, 1\}$. Agent 1 observes $o_1$ and sends message $m_1$. Agent 2 observes $o_2$ and receives $m_1$. Write the RIAL policy gradient $\nabla_\theta J$ for this game, expanding all terms in $\pi_i(a_i, m_i \mid o_i, m_{-i}^{\text{recv}})$ explicitly. Identify which terms contribute to learning the communication protocol.

**Exercise 10.3** (TarMAC Subsumes CommNet). ($**$)
Provide a complete proof that TarMAC (Definition 6.1) reduces to CommNet (Definition 4.1) when: (a) $\mathbf{v}_i = \mathbf{h}_i$ (identity value function), (b) $\mathbf{q}_j = \mathbf{q}$ for all $j$ (uniform query), and (c) $\mathbf{k}_i = \mathbf{k}$ for all $i$ (uniform key). Verify that conditions (b) and (c) imply uniform attention weights, and substitute into the TarMAC aggregation formula.

**Exercise 10.4** (DRU Gradient Analysis). ($**$)
For the DRU (Definition 5.2) with temperature $\tau$, compute:

(a) The gradient $\partial \text{DRU}_\tau(\tilde{m}) / \partial \tilde{m}$ during training for scalar $\tilde{m} \in \mathbb{R}$.

(b) Show that as $\tau \to 0^+$, the DRU output approaches the hard threshold $\mathbf{1}[\tilde{m} > 0]$ and the gradient approaches a scaled Dirac delta at $\tilde{m} = 0$.
(c) Compare the variance of the DIAL gradient (using the DRU) with the REINFORCE gradient for a single binary message, using a concrete example with a known reward function.

*Hint:* For part (b), use the fact that $\sigma(x/\tau) \to \mathbf{1}[x > 0]$ pointwise as $\tau \to 0^+$, and $\sigma'(x/\tau)/\tau \to \delta(x)$.

**Exercise 10.5** (Constructing a Dec-POMDP with VoC $= 0$)**.** $(***)$
Construct a Dec-POMDP with two agents, partial observability (neither agent observes the full state), and yet VoC $= 0$. Specifically:
(a) Define a state space $\mathcal{S}$, observation functions $O_1, O_2$, and reward function such that each agent's observations are incomplete but its optimal action depends only on its own observation.
(b) Prove formally that VoC $= 0$ by showing that condition (3) of Proposition 8.3 fails.
(c) Explain intuitively why partial observability alone is not sufficient for communication to help.

**Exercise 10.6** (Communication Protocol Design)**.** $(***)$
Consider three agents in a grid world. Agent 1 observes the target location $\ell \in \{1, \ldots, 9\}$ (one of 9 grid cells). Agents 2 and 3 observe nothing about the target. All agents can send messages from $\mathcal{M}_i = \{0, 1\}^b$ ($b$-bit messages).
(a) What is the minimum value of $b$ such that agent 1 can communicate the target location exactly? Justify using information-theoretic arguments.
(b) Design an explicit communication protocol (mapping from $\ell$ to messages) for the minimum $b$.
(c) Suppose communication is noisy: each bit is flipped independently with probability $p < 1/2$. Compute the probability of correct target identification as a function of $b$ and $p$, and determine the minimum $b$ to achieve error probability at most 0.01 for $p = 0.1$.

*Hint:* For (c), use repetition coding and apply Hoeffding's inequality.

## 11 Summary

- A *communication-augmented Dec-POMDP* extends the standard Dec-POMDP with message spaces, a communication protocol, and policies that condition on received messages as well as local observations (Definition 3.2).
- With sufficient bandwidth and broadcast topology, communication reduces the Dec-POMDP from NEXP-complete to PSPACE-complete by enabling centralized information aggregation (Proposition 3.4).
- **CommNet** achieves end-to-end differentiable communication by passing the mean of agents' hidden states as a continuous message, enabling standard backpropagation through the communication channel (Theorem 4.2).
- **RIAL** treats discrete messages as reinforcement learning actions, using REINFORCE-style gradients; **DIAL** uses a continuous relaxation (DRU) during training for lower-variance gradients, discretizing only at execution (Proposition 5.4).
- **TarMAC** introduces attention-based targeted communication, allowing each agent to selectively attend to the most relevant messages. TarMAC strictly generalizes CommNet (Proposition 6.2).

- *Emergent communication* arises when agents develop protocols from scratch via RL training; compositionality can be measured by topographic similarity (Definition 7.4).
- Non-trivial communication emerges under four necessary conditions: partial observability, complementary information, action interdependence, and sufficient channel capacity (Theorem 7.5).
- The *value of communication* VoC is positive if and only if agents have complementary observations and interdependent optimal actions (Proposition 8.3); achieving $\epsilon$-optimal performance requires $O(|\mathcal{S}| \log(T/\epsilon))$ bits per agent per step (Theorem 8.4).
- Key open problems include scalability to thousands of agents, adversarial robustness of communication, explicit communication cost modeling, and interpretability of learned protocols.

## 12 Further Reading

- **Phase 06 applications.** Lesson 66 (continuous control) and Lesson 70 (multi-vehicle coordination) apply communication-based MARL to aerospace scenarios, including formation control with limited bandwidth and communication-aware collision avoidance.
- **Learning when to communicate.** Singh et al. [2019] introduce IC3Net, which augments CommNet with a learned gating mechanism that decides at each time step whether an agent should communicate, significantly reducing bandwidth in cooperative-competitive settings.
- **Attentional communication.** Jiang and Lu [2018] propose ATOC, which uses an attention mechanism to form dynamic communication groups, and demonstrate scalability advantages in large-scale cooperative navigation.
- **Bidirectionally-coordinated nets.** Peng et al. [2017] develop BiCNet for StarCraft micromanagement, using bidirectional recurrent communication for real-time coordination among heterogeneous units.
- **Emergent language and NLP.** Lazaridou et al. [2017] study multi-agent cooperation and the emergence of natural language properties. The connection between emergent communication and computational pragmatics is surveyed in the broader NLP literature on language grounding.
- **Comprehensive surveys.** Zhang et al. [2021] provide a theoretical treatment of MARL including communication, while Hernandez-Leal et al. [2019] offer a critical perspective on deep MARL methods and their limitations.
- **Historical note.** The study of communication in multi-agent systems predates deep RL, with roots in distributed AI and the speech act theory of Austin and Searle. Shoham and Leyton-Brown [2008] cover the logical and game-theoretic foundations of multi-agent communication protocols.

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